Indic Speech-to-LLM Evaluation

Tamil and English ASR, noise robustness, and text-response generation

Prepared by: Vimal D

Final Report

This report consolidates the supplied Colab notebook, per-example CSV outputs, ASR metrics, noise experiment, LLM responses, and latency summary. It includes an exploratory qualitative review of 20 response examples; the review is explicitly labeled subjective and should not be treated as a gold-standard benchmark.

1. Abstract

This project evaluates an offline speech-to-text-to-LLM pipeline on 100 read-speech examples: 50 Tamil and 50 English samples from Google FLEURS. Whisper-small transcribes audio, then Qwen2.5-0.5B-Instruct generates a text response from each transcript. The ASR experiment compares beam search with five beams against greedy decoding with one beam, measures WER, CER, latency and real-time factor, and tests synthetic Gaussian noise at 15 dB SNR. Beam search improves Tamil WER substantially in this sample and slightly improves English WER. Synthetic noise increases WER and CER for both languages. An exploratory review of 20 generated responses found multiple incomplete or unclear Tamil responses and examples of unsupported or inaccurate additions in English responses. This sequential pipeline does not demonstrate streaming speech-to-speech interaction.

2. Objectives and setup

Objectives: (1) compare two recent speech-to-speech architectures relevant to low-latency dialogue; (2) build a Tamil/English audio-to-text-to-LLM pipeline; (3) evaluate ASR accuracy, runtime, and robustness under synthetic noise; and (4) propose a measurable low-latency improvement.

Dataset: Google FLEURS test split, 50 samples configured as `ta_in` and 50 as `en_us`. FLEURS contains read speech, not a purpose-built conversational QA benchmark. **Models:** `openai/whisper-small` and `Qwen/Qwen2.5-0.5B-Instruct`. **Runtime:** Google Colab, NVIDIA Tesla T4 GPU as reported. **Conditions:** `num_beams=5` versus `num_beams=1`; synthetic Gaussian noise at 15 dB SNR.

3. Pipeline and metrics

Audio (Tamil/English) → Whisper-small ASR → recognized transcript → Qwen2.5-0.5B-Instruct → text response → evaluation.

The implementation completes ASR before generating a text response. It does not produce speech audio or implement full-duplex turn-taking. WER and CER measure transcription errors (lower is better). RTF is processing time divided by audio duration; RTF below 1 indicates average processing faster than the audio duration, not proof of streaming performance.

4. ASR results: beam search vs greedy decoding

Language	Method	WER	CER	Mean latency (s)	Mean RTF
Tamil	baseline	0.7334	0.2602	2.4331	0.1876
Tamil	greedy	1.1134	0.4691	1.7439	0.1378
English	baseline	0.2507	0.0681	0.5154	0.0565
English	greedy	0.2545	0.0692	0.3382	0.0371

Tamil WER decreases from 1.1134 with greedy decoding to 0.7334 with beam search. English WER changes from 0.2545 to 0.2507. Beam search has higher measured latency than greedy decoding in both languages. WER can exceed 1.0 when word errors exceed the number of words in the reference; it should not be capped at 1.0.

5. Noise robustness

Language	Condition	WER	CER
Tamil	Clean	0.7334	0.2602
Tamil	Noisy (15 dB SNR)	0.8010	0.3324
English	Clean	0.2507	0.0681
English	Noisy (15 dB SNR)	0.2817	0.0902

At 15 dB synthetic Gaussian noise, Tamil WER increases from 0.7334 to 0.8010 and CER from 0.2602 to 0.3324. English WER increases from 0.2507 to 0.2817 and CER from 0.0681 to 0.0902. This suggests reduced accuracy under the tested noise condition, but does not establish performance for real traffic, crowd, reverberation, or competing-speaker noise.

6. End-to-end latency

Language	Samples	Mean ASR (s)	Mean LLM (s)	Mean end-to-end (s)
English	50	0.515	2.115	2.631
Tamil	50	2.433	3.345	5.778

Mean sequential end-to-end latency was 2.631 seconds for English and 5.778 seconds for Tamil. These are offline measurements; they are not time-to-first-token, streaming, microphone capture, or playback latency.

7. Representative outputs

Tamil — tamil_000

Reference: ■■■ ■■■■■■■■■■ ph ■■ ■■■■■■■■■■■■■■■■■■ ■■■■■■■■ ■■■■■■■■ ■■■■■■■■■■■■■ ■■■■■■■■■ ■■■■■■■■■■■■ ■■■■ ■■■■■■■■■■■■■ ■■■■■■■■■■■■■■

ASR prediction: ■■■■ ■■■■■■■■■ ■■■■■■■■■■ ■■■■■■■■■■■■■■■■■■■■■■ ■■■■■■■■ ■■■■■■■ ■■■■■■■■■■■■■ ■■■■■■■■■ ■■■■■■■■■■■■ ■■■■ ■■■■■■■■■■■■■ ■■■■■■■■■■■■■.

LLM response: ■■■■ ■■■■■■■■ ■■■■■■ ■■■■■■ ■■■■■■■■■ ■■■■■■■■■■ ■■■■■■■■■■■■■■■■■ ■■■■■■■ ■■■■■■■■■■■■■■ ■■■■■■■■■ ■■■■■■

English — english_000

Reference: however due to the slow communication channels styles in the west could lag behind by 25 to 30 year

ASR prediction: However, due to the slow communication channels, styles in the West could lag behind by 25 to 30 years.

LLM response: This statement highlights a significant gap between the rapid pace of technological advancements and cultural shifts in the Western world. Communication barriers can indeed lead to a slower pace of change compared to other regions or cultures. This disparity underscores the importance of bridging these gaps through improved infrastructure, better technology, and increased collaboration across borders.

The Tamil example is visibly incomplete at the response boundary. The English example is fluent but adds claims about technological progress, infrastructure, and cross-border collaboration that are not in the source statement. These examples illustrate why response fluency alone is not evidence of faithfulness.

8. Exploratory response-quality review

A qualitative review was performed on the first 10 outputs for each language in the supplied pipeline CSV. Labels are subjective and assistant-assisted, not independent expert annotations. `Faithful` means the response preserves the core meaning of the input; `Understandable` means the response is coherent enough to interpret; `Unsupported/inaccurate claim` flags at least one notable invented, contradictory, or unsupported detail. A score of 1 means the condition was flagged/present.

Language	N	Faithful	Understandable	Unsupported/inaccurate flagged
Tamil	10	2/10	1/10	8/10
English	10	3/10	10/10	9/10

In this exploratory sample, Tamil outputs were often incomplete or unclear. English outputs were generally more readable but frequently added unsupported or inaccurate details. These counts describe only the first 10 examples per language and must not be presented as rates for the full 100-sample dataset. A more rigorous evaluation should use independent annotators and a spoken QA dataset with verified reference answers.

9. Architecture comparison: Qwen2.5-Omni and Moshi

Qwen2.5-Omni. The technical report describes a multimodal model for text, audio, image and video input and text/audio output. Its Thinker–Talker design separates high-level perception and text reasoning from speech-token generation. Block-wise processing and streaming audio decoding are intended to reduce response delay [1].

Moshi. Moshi is designed for full-duplex speech dialogue. It uses discrete audio tokens from the Mimi neural codec and models user and assistant streams in parallel, with text aligned to speech to support natural turn-taking and interruptions [2].

Dimension	Qwen2.5-Omni	Moshi
Core architecture	Thinker–Talker; multimodal	Speech-text model; full duplex
Speech representation	Generated speech tokens	Mimi neural codec tokens
Streaming strategy	Block-wise processing; streaming decoding	Parallel user/assistant streams
Conversation emphasis	Multimodal input and output	Overlap, interruptions, turn-taking

The two architectures were not benchmarked directly in this project. Their reported latency values should not be compared without matching hardware, tasks and measurement methods.

10. Low-latency limitation and proposed improvement

A sequential ASR → LLM pipeline waits for transcription before response generation. ASR errors propagate to the LLM; long chunks, autoregressive generation, endpoint detection, buffering and language-dependent recognition quality can add delay. Full-duplex interaction also requires interruption handling and synchronization of incoming and outgoing streams.

Proposed improvement: implement latency-aware incremental processing with short overlapping audio chunks, partial ASR hypotheses, endpoint/confidence gating, and incremental LLM output. If the transcript is uncertain, ask for clarification rather than inventing content. Evaluate against the current baseline using WER/CER, time-to-first-transcript, time-to-first-response, end-to-end latency, interruption recovery and manual faithfulness review. This streaming design is proposed future work; it was not implemented or measured here.

11. Limitations and conclusion

Limitations: 50 samples per language; read speech rather than conversational QA; synthetic Gaussian noise rather than real-world recordings; one Colab hardware setup; no statistical significance analysis; no implemented streaming system; and a small subjective response review that should not be generalized to all outputs.

The prototype ran ASR and LLM response generation for 100 Tamil and English samples. Five-beam decoding improved Tamil WER substantially and English WER slightly relative to greedy decoding. Synthetic Gaussian noise degraded transcription metrics in both languages. Mean sequential end-to-end latency was 2.631 seconds for English and 5.778 seconds for Tamil. The qualitative review indicates that response faithfulness is an important next area for improvement. The results motivate future work on language-aware ASR, noise robustness, faithful response generation and incremental streaming.

References

[1] Xu et al. (2025). *Qwen2.5-Omni Technical Report*. https://arxiv.org/abs/2503.20215

[2] Défossez et al. (2024). *Moshi: a speech-text foundation model for real-time dialogue*. https://arxiv.org/abs/2410.00037

[3] Radford et al. (2022). *Robust Speech Recognition via Large-Scale Weak Supervision*. https://arxiv.org/abs/2212.04356

[4] Google FLEURS dataset: https://huggingface.co/datasets/google/fleurs

[5] Qwen2.5-0.5B-Instruct model card: https://huggingface.co/Qwen/Qwen2.5-0.5B-Instruct